SALARY SURVEY ANALYSIS

CREATED BY

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DADS JULY BATCH

MILESTONE PROJECT 3

OVER ALLVIEW

- Cleaned in Microsoft Excel
- SQL Queries for Analysis
- Tableau Dashboard for Salary Analysis

CLEANING IN EXCEL

- Standardize Data Types - Ensure all columns have appropriate data types
- Handle Missing Values – Removed columns which have 60% null values,Null values contains categorical columns are replaced with unknowns
- Handle Inconsistent Values – replaced the values which are inconsistent
- New columns – created new columns for the further analysis

SQL QUERIES FOR ANALYSIS

```sql
-- Average Salary by Industry and Gender
SELECT
    Industry,
    Gender,
    AVG(Total_Salary_with_Compensation) AS avg_salary
FROM salary_CLEANED
GROUP BY Industry, Gender
ORDER BY Industry, Gender;
```

Industry	Gender	avg_salary
academia	Woman	38000.0000
academia - stem	Woman	34000.0000
academia / research	Woman	68500.0000
academia--cell and molecular biology	Woman	48000.0000
academic medicine	Man	57500.0000
academic press production	Woman	44000.0000
academic publishing	Woman	50466.6667
academic research	Man	72750.0000

```sql
-- Total Salary Compensation by Job Title
SELECT
    Job_Title,
    SUM(Total_Salary_with_Compensation) AS total_salary_compensation
FROM salary_CLEANED
GROUP BY Job_Title
ORDER BY total_salary_compensation DESC;
```

Job_Title	total_salary_compensation
software engineer	58949536
senior software engineer	40909349
director	29951274
project manager	24390853
engineering manager	18379386
program manager	18338789
product manager	18092738
attorney	17116220

```sql
-- Number of Employees by Industry and Years of Experience
SELECT
    Industry,
    Years_of_Professional_Experience_Overall AS Experience_Level,
    COUNT(*) AS num_employees
FROM salary_cleaned
GROUP BY Industry, Years_of_Professional_Experience_Overall
ORDER BY Industry, num_employees DESC;
```

Industry	Experience_Level	num_employees
academia	2 - 4 years	1
academia - stem	2 - 4 years	1
academia / research	11 - 20 years	1
academia--cell and molecular biology	5-7 years	1
academic medicine	8 - 10 years	1
academic press production	8 - 10 years	1
academic publishing	8 - 10 years	2
academic publishing	11 - 20 years	2

Result 28

SQL

```sql
-- Median Salary by Age Range and Gender
SELECT Age_Range, Gender,
       Salary AS median_salary
FROM (
    SELECT Age_Range, Gender, Total_Salary_with_Compensation AS Salary,
           ROW_NUMBER() OVER (PARTITION BY Age_Range, Gender ORDER BY Total_Salary_with_Compensation) AS row_num,
           COUNT(*) OVER (PARTITION BY Age_Range, Gender) AS total_rows
    FROM salary_cleaned
) ranked
WHERE row_num = CEIL(total_rows / 2);
```

Age_Range	Gender	median_salary
18-24	Man	56000
18-24	Non-binary	39500
18-24	Prefer not to answer	52500
18-24	Woman	56000
25-34	Man	92000
25-34	Non-binary	57000
25-34	Prefer not to answer	75000
25-34	Woman	72500

```sql
-- Salary Distribution by Education Level
SELECT
    Highest_Level_of_Education_Completed AS Education_Level,
    AVG(Total_Salary_with_Compensation) AS avg_salary,
    MIN(Total_Salary_with_Compensation) AS min_salary,
    MAX(Total_Salary_with_Compensation) AS max_salary
FROM salary_CLEANED
GROUP BY Highest_Level_of_Education_Completed
ORDER BY avg_salary DESC;
```

Education_Level	avg_salary	min_salary	max_salary
Professional degree (MD, JD, etc.)	155684.1872	100	1900000
PhD	119616.5406	0	2002000
Master's degree	100164.7743	0	5004044
High School	97733.1496	49	5000040
College degree	95762.1103	0	3600000
Some college	86591.4197	0	1615000

```sql
-- Total Compensation by Job Title and Years of Experience
SELECT
    Job_Title,
    Years_of_Professional_Experience_Overall AS Experience_Level,
    SUM(Total_Salary_with_Compensation) AS total_compensation
FROM salary_cleaned
GROUP BY Job_Title, Years_of_Professional_Experience_Overall
ORDER BY total_compensation DESC;
```

Job_Title	Experience_Level	total_compensation
senior software engineer	11 - 20 years	19000158
director	11 - 20 years	15588254
software engineer	5-7 years	15291302
software engineer	11 - 20 years	13418150
software engineer	8 - 10 years	12817807
software engineer	2 - 4 years	12255716
project manager	11 - 20 years	9114690
senior software engineer	8 - 10 years	8301083

Result 30

```sql
-- Job Titles with the Highest Salary in Each Country

SELECT Country, Job_Title, Max_Salary
FROM (
    SELECT Country, Job_Title, Total_Salary_with_Compensation AS Max_Salary,
           RANK() OVER (PARTITION BY Country ORDER BY Total_Salary_with_Compensation DESC) AS salary_rank
    FROM salary_cleaned
) ranked
WHERE salary_rank = 1;
```

Country	Job_Title	Max_Salary
Argentina	full stack developer	116000
Australia	director engineer product management	380000
Belgium	compliance	230000
Canada	managing director	650000
Canada	software development manager	650000
China	student	5000040
France	cybersecurity principal	130000
Germany	cyber security management	1550000

```sql
-- Average Salary by City and Industry
SELECT
    City,
    Industry,
    AVG(Total_Salary_with_Compensation) AS avg_salary
FROM salary_cleaned
GROUP BY City, Industry
ORDER BY City, avg_salary DESC;
```

City	Industry	avg_salary
aberdeen	sales	40500.0000
aberdeen	health care	34300.0000
aberdeen	retail	33200.0000
adairsville	engineering or manufacturing	69940.0000
addis	engineering or manufacturing	87000.0000
addison	computing or tech	118250.0000
affluent ...	computing or tech	159000.0000
aiken	engineering or manufacturing	152500.0000

SQL

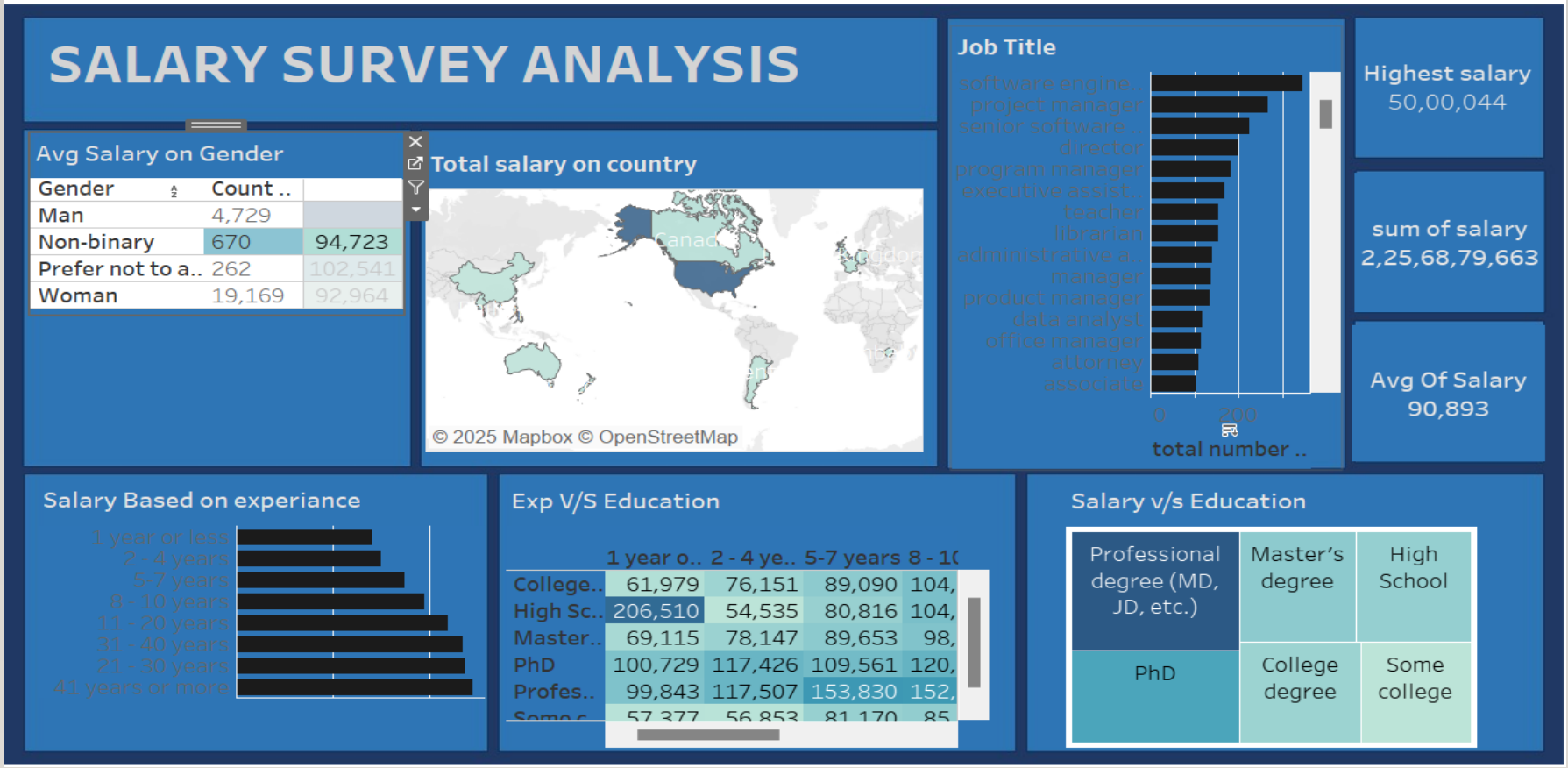
```sql
-- Percentage of Employees with Additional Monetary Compensation by Gender
SELECT
    Gender,
    COUNT(CASE WHEN Additional_Monetary_Compensation > 0 THEN 1 END) * 100.0 / COUNT(*) AS percentage_with_compensation
FROM salary_cleaned
GROUP BY Gender
ORDER BY percentage_with_compensation DESC;
```

Gender	percentage_with_compensation
Man	76.71812
Prefer not to answer	71.75573
Woman	70.87485
Non-binary	66.11940

```sql
-- Average Salary by Industry, Gender, and Education Level
SELECT
    Industry,
    Gender,
    Highest_Level_of_Education_Completed AS Education_Level,
    AVG(Total_Salary_with_Compensation) AS avg_salary
FROM salary_cleaned
GROUP BY Industry, Gender, Highest_Level_of_Education_Completed
ORDER BY Industry, Gender, avg_salary DESC;
```

Industry	Gender	Education_Level	avg_salary
academia	Woman	PhD	38000.0000
academia - stem	Woman	College degree	34000.0000
academia / research	Woman	Master's degree	68500.0000
academia--cell and molecular biology	Woman	PhD	48000.0000
academic medicine	Man	College degree	57500.0000
academic press production	Woman	Master's degree	44000.0000
academic publishing	Woman	College degree	64750.0000
academic publishing	Woman	Master's degree	43500.0000

TABLEAU DASHBOARD



INSIGHTS

- **Average Salary** - 90893
- **Total Salary** – 2,25,68,79,63
- **Total Employees Analyzed** – 24830
- United states of America earns the total highest salary
- The employees are more works in software engineer
- Most of the employees who hold professional degree are more
- Compared to men , women are more working
- By experience the employees who have 40+ years of experience earns the most